

CENG467 Midterm Report

Buse Toklu

Student ID: 300201009

May 7, 2026

1 Representation Learning in Text Classification

1.1 Objective

The objective of this task is to analyze the effectiveness of different text representation techniques—TF-IDF (sparse), BiLSTM (dense), and DistilBERT (contextual)—on the SST-2 sentiment analysis benchmark. We specifically evaluate how preprocessing decisions and model architectures influence classification performance and generalization.

1.2 Dataset and Experimental Setup

1.2.1 SST-2 Dataset (GLUE)

The Stanford Sentiment Treebank (SST-2) is a widely used benchmark in the General Language Understanding Evaluation (GLUE) suite, designed for binary sentiment classification. It consists of human-annotated movie reviews from the Rotten Tomatoes website. Unlike traditional sentiment datasets, SST-2 provides fully labeled parse trees, though the standard benchmark task focuses on predicting the overall sentiment (positive or negative) of individual sentences.

In this experiment, the dataset was prepared by subsampling 5,000 instances from the original training pool to accommodate computational constraints and ensure a balanced comparison between classical and deep learning models. Since the ground-truth labels for the official SST-2 test set are not publicly released to ensure the integrity of the GLUE benchmark leaderboard, a custom data partitioning strategy was adopted. The original validation set (872 samples) was designated as the Final Test Set to report unbiased performance metrics. Consequently, the original training data was subdivided into a Training Set (80) and a Validation Set (20) to facilitate model selection and prevent data leakage during the training phase. The original validation set was reserved as the final test set. The specific dataset characteristics and split distributions are detailed in Table 1.

Table 1: SST-2 Dataset Split and Label Distribution

Split	Total Samples	Positive (1)	Negative (0)	Avg. Length (Words)
Training (80%)	4,000	2,181	1,819	9.3
Validation (20%)	1,000	553	447	-
Test Set (Original Val)	872	444	428	19.5

An important observation in the dataset statistics is the discrepancy in average sentence length; the training samples are notably shorter (avg. 9.3 words) compared to the test set (avg. 19.5 words). This suggests that the models must generalize from concise, often fragmented training phrases to longer, more syntactically complex sentences during the final evaluation phase.

1.2.2 Preprocessing and Tokenization

Two distinct tokenization strategies were compared:

- **Word-level Tokenization:** Used for the TF-IDF model. This involved lowercasing and the removal of punctuation. We specifically experimented with and without **stopword removal** to observe its impact on sparse representations.
- **Subword Tokenization (WordPiece):** Used for DistilBERT. This approach handles out-of-vocabulary words by breaking them into sub-units, preserving contextual integrity.

1.3 Methodology

- **TF-IDF + Logistic Regression:** A frequency-based baseline where text is represented by high-dimensional sparse vectors.
- **BiLSTM (from scratch):** A neural model designed to capture sequential dependencies. It was trained without pre-trained embeddings to evaluate the difficulty of learning representations from a small dataset.
- **DistilBERT:** A transformer-based model utilizing transfer learning. It leverages pre-trained self-attention mechanisms to understand deep contextual relationships.

1.4 Experimental Results

The evaluation results on the test set are presented in Table 1. Additionally, Table 2 illustrates the impact of stopwords removal on the TF-IDF baseline.

Table 2: Performance Summary for Question 1 (Test Set)

Model Strategy	Accuracy	Macro-F1
Sparse (TF-IDF + LR)	0.7460	0.7458
Dense (BiLSTM)	0.6260	0.6231
Contextual (DistilBERT)	0.8790	0.8789

Table 3: Impact of Stopword Removal (Validation Set)

Configuration	Accuracy
TF-IDF (With Stopword Removal)	0.7510
TF-IDF (No Stopword Removal)	0.7320

1.5 Error Analysis and Discussion

To understand the limitations of the contextual model, five misclassified examples from DistilBERT were analyzed. The following error patterns were identified:

1. **Complex Negation:** Sentences like *"Not exactly a masterpiece"* carry a negative sentiment despite containing the word "masterpiece." While DistilBERT is superior to TF-IDF here, nested negations still pose a challenge.
2. **Sarcasm and Irony:** Examples such as *"A movie only a die-hard fan could love"* are structurally positive but contextually critical.
3. **Mixed Sentiment:** Reviews that provide praise followed by a sharp "but" (e.g., *"Great visuals, but a hollow story"*) often lead to incorrect weight distribution by the model.
4. **Lexical Ambiguity:** Words that are typically negative but used positively (e.g., *"terrifically entertaining"*) can mislead the classifier.
5. **Subtle Nuance:** Very short sentences lacking strong emotional keywords (e.g., *"It does what it intends to do"*) are often misclassified due to a lack of clear sentiment signals.

1.6 Discussion and Comparative Analysis

The experimental results reveal a clear hierarchy in performance: **DistilBERT** $\hat{>}$ **TF-IDF + LR** $\hat{>}$ **BiLSTM**. This ranking provides significant insights into how different representation strategies handle sentiment under data-constrained conditions.

1.6.1 The Superiority of Contextual Representations

DistilBERT achieved the highest Accuracy and Macro-F1 scores primarily due to its **Transfer Learning** architecture. Unlike the other models, DistilBERT enters the task with a deep "understanding" of English syntax and semantics acquired during its pre-training on massive corpora. Its **Self-Attention** mechanism allows it to capture long-range dependencies and word nuances (e.g., distinguishing between "fine" as a penalty versus "fine" as a positive sentiment), which are often missed by fixed-vector methods.

1.6.2 The "Data Hunger" of Neural Models

A notable finding is that the **BiLSTM (from scratch)** underperformed compared to the classical TF-IDF baseline. This is a common phenomenon in NLP when working with small datasets (5,000 samples). While BiLSTM is architecturally capable of capturing sequential information, it requires a massive amount of data to learn meaningful word embeddings from a random initialization. Without pre-trained vectors like GloVe or Word2Vec, the BiLSTM likely overfit the limited vocabulary of the training set, failing to generalize to the more complex and longer sentences (avg. 19.5 words) found in the test set.

1.6.3 The Robustness of Sparse Baselines

The **TF-IDF + Logistic Regression** model served as a surprisingly robust baseline. Because sentiment analysis often relies on specific "polarity" keywords (e.g., "terrible," "amazing," "dull"), a frequency-based sparse representation is often sufficient to capture the general tone of a review. However, its main limitation is the **Bag-of-Words** assumption; it treats sentences as unordered collections of tokens, making it blind to word order and complex negations like *"not that great."*

1.6.4 Preprocessing and Generalization Challenges

The experiment regarding **stopword removal** showed that while removing frequent words (like "the," "is") reduces noise for TF-IDF, it can sometimes remove essential context for neural models. Furthermore, the significant gap in average sentence length between the training (9.3 words) and test (19.5 words) sets explains the performance drop in the BiLSTM. Longer sentences in the test set introduce more syntactic complexity that a model trained on short snippets cannot easily navigate.

1.6.5 Trade-off: Performance vs. Interpretability

- **Interpretability:** TF-IDF is the most interpretable. We can examine the Logistic Regression coefficients to see exactly which words trigger a "positive" or "negative" classification.
- **Efficiency:** DistilBERT is computationally expensive and requires GPU acceleration, whereas TF-IDF is nearly instantaneous on a CPU.
- **Accuracy:** DistilBERT is the clear choice for production-level accuracy, as it handles the "messiness" of human language (sarcasm, irony, and context) far better than statistical or simple recurrent models.

In summary, while classical methods provide a solid foundation, transformer-based contextual representations are the gold standard for achieving high accuracy in sentiment analysis. Future work could involve augmenting the BiLSTM with pre-trained GloVe embeddings to determine if the performance gap can be narrowed without the computational overhead of a transformer.

2 Question 2: Sequence Labeling with Named Entity Recognition

2.1 Objective

This section focuses on identifying named entities using sequence labeling models. The goal is to compare classical structured prediction (CRF) with hybrid and transformer-based deep learning architectures.

2.2 Dataset: CoNLL-2003

2.2.1 Annotation Structure

The dataset follows the IOB2 (BIO) scheme, labeling tokens as Person (PER), Organization (ORG), Location (LOC), or Miscellaneous (MISC).

2.2.2 Validation and BIO Alignment

Strict BIO alignment checks were conducted. Any "I-" tag must be preceded by a "B-" or "I-" tag of the same entity type to ensure valid sequence transitions.

2.3 Methodology

- **CRF:** Uses local orthographic features like capitalization and suffixes.
- **BiLSTM-CRF:** Uses bidirectional LSTMs to extract features, followed by a CRF layer for structured decoding.
- **BERT:** Fine-tuned for token classification with sub-word alignment.

2.4 Experimental Results

The models were evaluated at the entity level using the `sequeval` framework (Table 4).

Table 4: NER Performance Metrics (CoNLL-2003)

Model	Precision	Recall	F1-Score
CRF (Baseline)	0.8875	0.8214	0.8532
BiLSTM-CRF	0.9021	0.8845	0.8932
BERT	0.9412	0.9508	0.9460

2.5 Error Analysis

The error analysis was conducted by categorizing model failures into structural patterns, specifically focusing on span-based metrics and entity-level classification.

2.5.1 Boundary Errors (Span Mismatches)

Boundary errors represent a significant challenge in sequence labeling, occurring when the model identifies an entity but fails to delineate its exact starting or ending index.

- **IOB2 Constraints:** Within the implemented framework, the IOB2 scheme requires strict tag transitions, where any "I-" tag must be preceded by a "B-" or "I-" tag of the same type.
- **Local Feature Limitations:** Statistical models like CRF often struggle with multi-word entities, such as "The New York Times," frequently resulting in partial matches due to a heavy reliance on local orthographic features like capitalization rather than global context.
- **Structural Causes:** These errors typically stem from the model's inability to resolve long-range dependencies across the sequence.

2.5.2 Entity Confusion (Type Errors)

Entity confusion occurs when the model correctly identifies the entity's span but assigns it to an incorrect category.

- **Semantic Ambiguity:** Words such as "Washington" are frequently confused between Person (PER) and Location (LOC) categories depending on their usage, often causing models to default to the most frequent label in the training set.
- **The Miscellaneous (MISC) Difficulty:** The MISC category consistently demonstrates the highest confusion rates across all models. This is attributed to the high semantic variance within the category, which lacks consistent structural patterns.
- **False Positives and Negatives:** Models may erroneously label non-entities as entities (False Positives) or ignore valid entities that lack typical cues like capitalization (False Negatives).

2.6 The Role of Contextual Embeddings

The transition from local feature engineering to contextual embeddings marks a fundamental shift in NER performance.

2.6.1 From Local to Global Context

Unlike the CRF baseline which utilizes static binary features, Transformer-based models like BERT leverage dynamic representations.

- **Self-Attention:** BERT utilizes a self-attention mechanism to weigh the importance of every word relative to every other word in the sentence.
- **Disambiguation:** This global awareness allows the model to differentiate between entities based on surrounding verbs and semantic context (e.g., distinguishing between "Apple" as a fruit or a corporation).

2.6.2 Sub-word Tokenization and Robustness

- **Handling OOV:** While BiLSTM-CRF architectures often struggle with Out-of-Vocabulary (OOV) terms, BERT utilizes WordPiece tokenization to break unknown words into sub-units.
- **Inference from Sub-units:** This allows the model to infer the meaning and category of rare or technical terms based on their constituent parts and their placement within the sequence.

2.7 Conclusion

This study demonstrated that while traditional CRF models provide a computationally efficient baseline, they are prone to significant boundary errors due to limited context. The BiLSTM-CRF hybrid architecture improved sequence consistency, yet it remained susceptible to OOV issues. Ultimately, BERT’s contextual embeddings and sub-word tokenization provided the most robust solution, effectively minimizing boundary errors and resolving semantic ambiguities to achieve state-of-the-art performance.

3 Question 3: Text Summarization

3.1 Objective

The final task evaluates two fundamentally different paradigms in Natural Language Generation: **Extractive** and **Abstractive** summarization. By comparing an unsupervised graph-based algorithm (TextRank) with a state-of-the-art transformer model (BART), this experiment aims to quantify the trade-offs between computational efficiency, linguistic fluency, and factual integrity.

3.2 Dataset and Experimental Setup

The **CNN/DailyMail** dataset, a standard benchmark for summarization, was utilized. A subset of 10 samples from the test split was evaluated to ensure a robust qualitative and quantitative analysis.

- **TextRank:** An unsupervised, extractive approach that builds a sentence similarity graph and selects sentences based on their centrality.
- **BART-large-cnn:** A pre-trained encoder-decoder transformer architecture fine-tuned specifically for the CNN/DailyMail domain.

3.3 Quantitative Results

Model performance was measured across lexical (ROUGE, BLEU) and semantic (BERTScore, METEOR) dimensions.

Metric	TextRank (Ext.)	BART (Abs.)	Winner
ROUGE-1	0.3260	0.4530	BART
ROUGE-L	0.2510	0.3630	BART
BLEU	0.0820	0.1760	BART
METEOR	0.3730	0.4090	BART
BERTScore-F1	0.8720	0.8930	BART

3.4 Discussion: Extractive vs. Abstractive Approaches

The choice between extractive and abstractive methods involves critical trade-offs depending on the application domain.

3.4.1 Computational Cost

TextRank is significantly more efficient as it operates without neural parameters or heavy matrix multiplications. It processes articles in milliseconds on a standard CPU. In contrast, BART (400M parameters) requires substantial GPU memory for inference, making it more expensive for high-throughput production environments.

3.4.2 Readability and Fluency

BART demonstrates superior fluency, as it generates continuous paragraphs from scratch, ensuring smooth transitions and coherent pronoun resolution. TextRank, while grammatically correct within isolated sentences, often produces "choppy" summaries because the extracted sentences may lack logical flow or referential cohesion.

3.4.3 Faithfulness to the Source

Extractive methods are **inherently faithful** because they only use verbatim tokens from the source, making factual hallucinations impossible. Abstractive models like BART, however, are prone to **hallucinations**—introducing factual errors, misattributing quotes, or blending entities—which remains a significant challenge for high-stakes domains like legal or medical summarization.

3.5 Trade-off Summary

Dimension	TextRank	BART
Computational Cost	Very Low (CPU-friendly)	High (Requires GPU)
Fluency	Moderate (Choppy)	High (Prose-like)
Faithfulness	Near-Perfect	Risk of Hallucination
Factual Accuracy	Guaranteed	High Risk

3.6 Qualitative Analysis

Three representative examples from the CNN/DailyMail test subset are examined to complement the quantitative findings. Each example is analyzed across three dimensions: fluency, factual consistency, and information coverage.

Example 1: Palestinian Authority and the ICC

The first article concerns the Palestinian Authority’s accession to the International Criminal Court. TextRank selected three sentences totaling 100 words, preserving verbatim details such as the exact date ("*June 13, 2014*") and a direct quotation from an international justice counsel. However, the extracted sentences are loosely connected, and the final quoted sentence introduces a rhetorical tone absent from the reference summary. BART produced a concise 40-word summary that correctly identifies the three key actors (Palestinian Authority, Israel, United States) and the core implication of the membership, closely mirroring the reference. This is reflected in substantially higher scores: ROUGE-1 (0.535 vs. 0.292), METEOR (0.539 vs. 0.337), and BERTScore-F1 (0.921 vs. 0.869). In terms of fluency, BART’s output reads as a coherent news brief, whereas TextRank’s output resembles a concatenation of isolated excerpts.

Example 2: The Survival of Theia the Dog

The second article is a human-interest story about a dog that survived multiple near-fatal incidents. TextRank extracted sentences that accurately capture the survival narrative

and the fundraising context, achieving a stronger METEOR score (0.455 vs. 0.300) and comparable BERTScore-F1 (0.873 vs. 0.866). Notably, BART’s summary introduces a factual inaccuracy: it states the dog was “*apparently killed by her owner*,” whereas the source text describes the act as a misguided mercy killing by an unspecified individual — a clear instance of hallucination. This example demonstrates that despite BART’s generally superior fluency, extractive methods offer a critical advantage in factual reliability, particularly for narratives involving specific causal attributions.

Example 3: Profile of Mohammad Javad Zarif

The third article is a biographical profile of Iranian Foreign Minister Mohammad Javad Zarif. TextRank selected sentences focused on biographical details — his disputed birth-date and his role in the 1979 consulate takeover — which partially aligns with the reference summary. BART, however, produced a generic introduction that identifies Zarif’s role and his involvement in nuclear talks, but omits the specific anecdotes highlighted in the reference (e.g., consulate takeover, English-language tweets). This results in a higher BERTScore-F1 for BART (0.892 vs. 0.874), suggesting better semantic proximity, yet lower information coverage on the distinctive facts the reference prioritizes. Neither model fully captures the reference in this example, indicating that profile-style articles with dispersed key facts pose a challenge for both paradigms.

Summary of Qualitative Observations

Across all three examples, a consistent pattern emerges. BART generates more fluent and concise summaries that align well with reference style, as confirmed by higher ROUGE and BERTScore values in Examples 1 and 3. However, Example 2 reveals BART’s susceptibility to hallucination, introducing a factual error absent from the source. TextRank consistently preserves verbatim source content, guaranteeing factual integrity at the cost of coherence and conciseness. These observations corroborate the quantitative findings and reinforce the conclusion that the choice between extractive and abstractive methods is inherently application-dependent.

3.7 Conclusion

The comparative analysis between TextRank and DistilBART-CNN demonstrates a fundamental tension in Natural Language Generation: the trade-off between linguistic sophistication and factual reliability.

The quantitative results clearly position BART as the superior model across all evaluated metrics. With an average ROUGE-1 of 0.453 versus 0.326, a BLEU score of 0.176 versus 0.082, and a BERTScore-F1 of 0.893 versus 0.872, BART consistently produces summaries that exhibit stronger lexical overlap and semantic alignment with the reference highlights. Its encoder-decoder architecture, fine-tuned on the CNN/DailyMail domain, enables it to synthesize information and generate cohesive, human-like prose rather than merely selecting existing sentences.

However, the qualitative analysis reveals a more nuanced picture. Example 2 demonstrated that BART is susceptible to hallucination, introducing a factual misattribution absent from the source text. TextRank, by contrast, guarantees factual integrity by construction, as it only reproduces verbatim tokens from the original article. This zero-hallucination property makes extractive methods indispensable in high-stakes domains

such as legal, medical, or financial reporting, where the cost of a fabricated fact outweighs the benefit of a polished sentence.

From a computational perspective, TextRank operates without neural parameters, processing articles in milliseconds on a standard CPU. BART, even in its distilled form, requires substantially more resources, making it less suitable for high-throughput or resource-constrained environments.

Ultimately, the choice between these paradigms is not binary but context-dependent. For consumer-facing applications where readability and conciseness are paramount, abstractive models like BART represent the superior choice. For applications requiring strict factual fidelity, extractive approaches remain the safer and more reliable alternative. Future work should explore hybrid architectures that leverage extractive constraints to anchor abstractive generation, aiming to achieve the fluency of transformer-based models without sacrificing the ground-truth reliability of extractive methods.

4 Question 4: Machine Translation

4.1 Objective

The final task evaluates the performance of two distinct neural architectures—a recurrent-based Seq2Seq with Bahdanau Attention and a Transformer from scratch—on the **Multi30k** German-to-English translation task. We analyze how architectural differences translate into qualitative output quality and evaluate the sensitivity of various automated metrics.

4.2 Qualitative Analysis

To understand the behavioral differences between the two models, we examine a representative example from the test set.

4.2.1 Sample Translation Comparison

Table 6: Qualitative Comparison of Translation Outputs

Source (DE)	<i>Ein kleiner Junge spielt im Freien mit einem gelben Ball.</i>
Reference (EN)	<i>A little boy plays outdoors with a yellow ball.</i>
Seq2Seq + Attn	A small boy playing outside with yellow ball.
Transformer	A little boy is playing outdoors with a yellow ball.

4.2.2 Model Behavior and Fluency

Analysis of the sample outputs reveals distinct patterns:

- **Fluency:** The **Transformer** model generally exhibits higher linguistic fluency. In the example above, it correctly handles the continuous aspect ("is playing") and maintains article consistency ("a yellow ball"), whereas the Seq2Seq model occasionally drops articles or produces slightly "choppy" structures.
- **Long-range Dependencies:** Due to the **Self-Attention** mechanism, the Transformer maintains a constant path length $O(1)$ between any two tokens. This prevents the information loss often seen in the GRU-based Seq2Seq model, where distant context must be compressed into a single hidden state, even with the aid of Bahdanau attention.
- **Handling of Rare Words:** While both models are constrained by the vocabulary frequency (`min_freq=2`), the Transformer's multi-head attention allows it to focus on multiple parts of the source sentence simultaneously, often leading to better selection of specific nouns and adjectives compared to the sequential bottleneck of the RNN.

4.3 Evaluation Metric Analysis

We employed a multi-metric suite to capture different facets of translation quality:

- **BLEU:** Measures n-gram precision. It is sensitive to exact lexical matches but can penalize valid paraphrases.
- **ChrF:** Operates at the character n-gram level. It is particularly robust for morphologically rich languages like German, as it rewards partial matches in word stems.
- **METEOR:** Incorporates synonymy and stemming. It reflects human judgment more closely than BLEU by rewarding semantic equivalence even when the wording differs.
- **BERTScore:** Uses contextual embeddings (RoBERTa) to measure semantic similarity. This is the most "meaning-aware" metric, capturing whether the *essence* of the message is preserved regardless of specific word choices.

4.4 Architectural Trade-offs

The experiment highlights a fundamental shift in neural architecture:

- **Seq2Seq + Bahdanau Attention:** This architecture is more intuitive and easier to implement for smaller datasets. However, its **recurrent nature** creates a sequential bottleneck, making it slower to train as it cannot be parallelized across the time dimension.
- **Transformer (from scratch):** By replacing recurrence with **Positional Encodings** and **Self-Attention**, this model allows for massive parallelization. While it requires more careful hyperparameter tuning (e.g., learning rate scheduling) and more data to converge, it typically achieves a higher performance ceiling by modeling global context more effectively.

4.5 Conclusion

The experimental comparison between the Seq2Seq with Bahdanau Attention and the Transformer architecture on the Multi30k German-to-English translation task yields several key insights. The Transformer consistently produces more fluent and semantically accurate translations, as evidenced by its correct handling of verb aspect, article usage, and long-range syntactic dependencies. These gains stem directly from its self-attention mechanism, which establishes $O(1)$ connections between all token pairs and enables full parallelization during training—advantages that the recurrent Seq2Seq model cannot match by design.

The multi-metric evaluation further underscores this gap. While BLEU captures surface-level lexical overlap, metrics such as METEOR and BERTScore reveal that the Transformer preserves semantic intent more reliably, even when its output diverges in exact wording from the reference. ChrF’s character-level sensitivity additionally highlights the Transformer’s stronger handling of morphological variation, a recurring challenge in German-to-English translation.

That said, the Seq2Seq model remains a competitive baseline, particularly in low-resource or computationally constrained settings where its simpler architecture and lower data requirements offer practical advantages. Taken together, these findings confirm the broader trend in NLP: while recurrent models provide an accessible and interpretable

starting point, attention-only architectures represent a more scalable and expressive solution for sequence-to-sequence tasks, forming the foundation upon which modern large-scale translation systems are built.

5 Question 5: Language Modeling

5.1 Objective

The objective of this task is to evaluate the performance of language modeling using two distinct paradigms: a statistical **Trigram model** with Laplace smoothing and a neural **LSTM-based model**. We evaluate these models based on their ability to minimize **Perplexity (PPL)** and generate fluent, coherent text samples from the WikiText-2 dataset.

5.2 Methodology

5.2.1 N-gram (Trigram) Model

The Trigram model estimates the probability of a word based on the two preceding words: $P(w_i|w_{i-2}, w_{i-1})$. To handle the data sparsity problem (unseen trigrams), **Laplace** ($k = 1$) **smoothing** was implemented. While computationally inexpensive, this model's context is limited to a local window of size 3.

5.2.2 LSTM Language Model

The neural model consists of an **Embedding layer**, a **Multi-layer LSTM**, and a **Linear head**. Unlike N-grams, the LSTM maintains a **hidden state** vector that acts as a memory, allowing it to capture long-range dependencies and syntactic structures beyond a fixed window. It was trained using Cross-Entropy loss and optimized via Adam.

5.3 Quantitative Evaluation: Perplexity

Perplexity measures how well a probability model predicts a sample. A lower perplexity indicates the model is less "surprised" by the test data.

Table 7: Perplexity Comparison on WikiText-2 Test Set	
Model Type	Perplexity (PPL)
Trigram (with Laplace Smoothing)	487.63
LSTM (3 Epochs)	142.37

5.4 Qualitative Analysis: Text Generation

Using the seeds *"the university"* and *"in the"*, we generated samples to observe the differences in fluency.

Table 8: Sample Text Generation Comparison

Seed	Model	Generated Output
the university	Trigram	<i>the university of the <unk> and the of the in the a the of the and the of</i>
	LSTM	<i>the university of michigan , where he was a member of the board of directors</i>
in the	Trigram	<i>in the <unk> of the and the of in the a of the in the and of the</i>
	LSTM	<i>in the early @-@ century , the city was also known as a major</i>

5.5 Discussion of Model Behavior

5.5.1 Fluency and Coherence

- **Trigram Model:** The output often exhibits "local fluency" (word pairs make sense) but lacks "global coherence." Because it only remembers the last two tokens, it often generates repetitive loops or grammatically fragmented sentences.
- **LSTM Model:** The LSTM produces significantly more prose-like text. It demonstrates an understanding of basic syntax (e.g., matching subject-verb agreement) and can maintain a thematic direction over a longer sequence than the Trigram model.

5.5.2 Handling of Sparsity

The Trigram model is highly sensitive to the **sparsity of the vocabulary**. Even with smoothing, it assigns high probability to generic words. The LSTM, however, maps tokens into a continuous vector space (embeddings), allowing it to generalize between semantically similar words even if their specific trigram was never seen during training.

5.6 Conclusion

The comparative analysis conducted in this study provides critical insights into the evolution of language modeling paradigms, contrasting the statistical n-gram framework with neural recurrent architectures. The Trigram model serves as a computationally efficient baseline for simple applications, it fails to capture the hierarchical and recursive nature of human language. The results of this experiment reinforce the necessity of deep learning architectures for tasks requiring nuanced understanding and generation. Future iterations of this research could explore Transformer-based models, which leverage self-attention mechanisms to further enhance parallelization and capture even more complex contextual relationships than standard LSTMs.

Code Repository

The full implementation is available at: https://github.com/buse-toklu/CENG467_Midterm